

What's new

Version 2.03

Support for Unity 6.2.0b1 and URP 17.2.0.

Version 2.03 has been tested in Unity 6.2.0b1 using URP 17.2.0. It adds compatibility for this version of URP.

2.03 is only available if you update the package using Unity 6000.2.0b1 or above.

Version 2.01

Support for Unity 6.0.45f1 and URP 17.0.4.

Version 2.01 has been tested in Unity 6.0.45f1 using URP 17.04. It optimizes the deferred lighting pass when using Lux URP Essential HLSL shaders which implement custom lighting by marking the related pixels in the stencil buffer accordingly. Examples are Skin, Hair or the Cloth shaders. Shader graph shaders already had this optimization built in.

2.01 is only available if you update the package using Unity 6000.0.45f1 or above.

Version 2.0

Support for Unity 6.0.23f1 and URP 17.0.3.

Version 2.0 adds support for Unity 6000.0.23f1 and URP 17.0.3. (October 11, 2024) and is only available if you update the package using Unity 6000.0.23f1

Version 1.9.9.d preview

Support for Unity 6.0.12f1 and URP 17.0.3.

Version 1.9.9.d adds support for Unity 6000.0.12f1 and URP 17.0.3. (July 17, 2024) and is only available if you update the package using Unity 6000.0.12f1

Changes and Fixes 1.9.9.d

- Fixed **motion vectors** in the **foliage shader** and added a checkbox to toggle motion vectors from vertex animations.

Known Issues

- The **_CameraDepthTexture** is not bound when using deferred rendering. So all shaders which rely on it will look odd (water, decals, particles). (Unity bug).
- **Light map preview on the terrain** is broken if terrain instancing is enabled. (Unity bug).

Version 1.9.9.c preview

Support for Unity 6.0.11f1 and URP 17.0.3.

Version 1.9.9.c adds support for Unity 6000.0.11f1 and URP 17.0.3. (July 11, 2024) and is only available if you update the package using Unity 6000.0.11f1

Changes and Fixes 1.9.9.c

- Fixed **debug view** in shader graph shaders using custom lighting functions.
- Added proper **APV lighting** to the **terrain** base shader.
- Rework the old, deprecated **terrain blending** and added a shader graph based alternative.
- Custom Function: **Detail texturing** added.

Known Issues

- All shaders (including the built in URP shaders) will **error** when using the rendering debugger and APV enabled. (Unity bug).
- The **_CameraDepthTexture** is not bound when using deferred rendering. So all shaders which rely on it will look odd (water, decals, particles). (Unity bug).
- **Light map preview on the terrain** is broken if terrain instancing is enabled. (Unity bug).

*The reworked **terrain blending** shader and the shader graph based alternative has been added to URP 14.0.10 as well with Version 1.98.5 for Unity 2022.3.20+*

Version 1.9.9.b preview

Support for Unity 6.0.9f1 and URP 17.0.3.

Version 1.9.9.b adds support for Unity 6000.0.9f1 and URP 17.0.3. (June 26, 2024) and is only available if you update the package using Unity 6000.0.9f1

Although the version of URP in 6000.0.8f1 and 6000.0.9f1 claims to be the same (17.0.3) it is not and 6000.0.9f1 added some fundamental changes which forced me to change a lot of shader code.

Changes and Fixes

- Adjusted the **GI lighting calls** so they work in 6000.0.9f1 and URP 17.0.3. (June 26, 2024)
- Fixed **APV sampling** in custom lighting functions for Shader Graph.
- Fixed **light map UVs** in shader graph shaders.

Known Issues

- All shaders (including the built in URP shaders) will **error** when using the rendering debugger and APV enabled. (Unity bug).
- The **_CameraDepthTexture** is not bound when using deferred rendering. So all shaders which rely on it will look odd (water, decals, particles). (Unity bug).
- **Light map preview on the terrain** is broken if terrain instancing is enabled. (Unity bug).

Version 1.9.9.a preview

Support for Unity 6.0.8f1 and URP 17.0.3.

Version 1.9.9.a adds support for URP 17.0.3. (preview) and is only available if you update the package using Unity 6000.0.8f1

Deprecated

Changes and Fixes

- Support for the **GPU Resident Drawer** added.
This broke a lot of old rendering solutions like the toon outlines. These now are rendered using a *Renderer Feature*. [Please refer to the "Lux URP Toon Shading 1.3" pdf to find out more.](#)
- Support for **APVs** added.
- Support for **Vertex Ambient Lighting** added to all Shader Graph shaders using custom lighting.
- **Motion Vectors** added.
At least I started to add them to most HLSL shaders. Shader Graph shaders should get them right out of the box. Quite tricky when it comes to the Uber shader and alpha testing plus parallax tho :) Foliage now supports proper per object motion vectors taking bending into account. Grass does not currently as it might just be too expensive.
- **Grass Displacement** has been rewritten and now uses the Render Graph.

Version 1.9.8.

Support for URP 16.0.4.

Version 1.98 adds support for URP 16.0.4. if you update the package using Unity 2023.2.0f1+

Deprecated

Changes and Fixes

- **Folder structure** updated
- Minor fixes for Metal added

- **Toon & Outline shader** Batching fixed

Version 1.9.6.

Support for URP 15.0.6.

Version 1.96 adds support for URP 15.0.6. if you update the package using Unity 2023.1.5f1.

Deprecated

Changes and Fixes

- **Terrain Shader** fixed procedural texturing when switching the build platform to Android.
- Increased lighting BRDF specular max for half float math (mobile) to match the visual look of full float math (desktop) better.
- Version 1.9.6b adds an experimental Shader graph function for "Best Fitting Normals".

Version 1.9.

Support for URP 14.0.4.

Version 1.9 adds support for URP 14.0.4. if you update the package using Unity 2022.2 and implements Forward+ lighting, rendering layers*, LOD cross fading and DOTS instancing to all the HLSL shader. Shader graphs will be updated automatically by Unity.

URP 14.0.4. still handles fog on transparents in a very special way. So I added the new blend modes but tweaked them to make them more pleasant. [See the "Fog on Transparents" demo scene.](#)

** Rendering layers however seem to be broken in URP: Even using the built in Lit shader they seem to only work when using deferred.*

Please note: You have to update the package using Unity 2022.2. to get the 14.0.4. shaders.

New nodes for Shader Graph

- **Fast top down projection** like used in the HLSL shader.
- **In shader tone mapping** locally or globally controlled using ACES, neutral tone mapping and custom settings like gamma, contrast and saturation
- **Standard lighting node** as it is needed by the in shader tone mapping
- **GPU instanced mesh particles** node.

Changes and Fixes

- **Skin Lighting node for Shader Graph** now lets you provide custom diffuse and specular normal samples as input enabling you to create some kind of simple wrinkle maps. [See the "Custom Shader Graphs Demo" for an example.](#)
- **Uber Shader GUI** fixed: It now lets you tweak the rendering queue again.

- **Terrain Mesh Blend shader** Altho marked as deprecated I had another quick look into this one and added some tweaks. The "Terrain Blend Depth Only" shader now does a full GBuffer pass when using deferred lighting while the "Terrain Blend" shader only adds the blending around the intersection with the terrain. SSAO now looks nicer but the shader still breaks if depth priming is active. Only solution is to set it to Render Queue >= Transparent.
- **Lit Particles** are flagged as deprecated now as well as their per vertex shadows do not work with Forward+. Shaders will just fall back to per pixel shadows at least for all additional lights. So the tessellation shader only produces smooth shadows for the directional light.
- **Lux URP Grass TextureDisplace shader** has been removed from the "Lux SRP Grass Displacement" folder: The default grass shader has an option to apply grass displacement.
- **Tree creator shaders** As all shaders using URP 14.0.4. now support LOD cross fading we can use tree creator trees like speed trees using LOD groups. So i added a simple billboard shader which supports geometry billboards based on a cross. *Check out the "Tree Creator LOD Demo" if you are on URP 14.0.4. This gives you real time lit and shadow casting billboards.*
- **Simple procedural skybox shader** added which supports fog. *Example can be found in teh "URP12+ Demo" scene.*
- **Overall cleanup** which fixed missing pass names, srp batcher being broken and things like this.

URP 14.0.4. broken features

URP is evolving but some of its features just break... Following issues are caused by URP and are not related to Lux URP Essentials as far as i can say:

- **Rendering Layers** are totally broken unless you use deferred lighting.
- **Deferred and Decals** are broken when using the Decal Buffer.
- **Directional shadows** on forward rendered materials such as skin or cloth look pixelated when using deferred rendering: Using the Lit Complex shader I get the same results as on the skin shader e.g. (regression in URP 14.0.4).
- **Additional light shadows** on forward rendered materials do not get rendered at all when using deferred rendering:
URP simply never defines `_ADDITIONAL_LIGHT_SHADOWS`

So if you do not plan to make heavy use of Forward+ or use deferred lighting anyway i recommend to stick with URP 12.1. or 13.1.8 for now.

Changes in older versions are tracked in the Changelog.txt only.